

A VERTICAL ELECTRICAL SOUNDING AND SELF-POTENTIAL METHODS TO SURVEY FOR PLACEMENT OF POTABLE WATER WELLS

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Abstract

Water is a precious commodity in most urban and rural areas. Lack of local potable water sources threatens not only thriving but a mere survival of rural communities all over the world. Establishing potable water wells requires a lot of fundings and resources and often cost prohibitive for local governments in South America and Africa. Searching for shallow groundwater require knowledge of subsurface layers and locating intensity and directions of water fluxes, which can be accomplished with geophysical methods of vertical electrical sounding (VES) and self-potential (SP). A method of VES can distinguish differences in electrical resistivity or conductivity at the multiple (10+) layers in soil profiles. These differences reveal the changes in soil texture and structure between water-bearing and waterproof layers, which form a framework for the subsurface water fluxes. The directions and intensities of the fluxes can then be evaluated with the self-potential method. However, conventional equipment for VES and SP is very expensive, bulky and complicated to operate. We tested a simple low-cost handheld device, LandMapper ERM-02, to evaluate layers in the ground with VES method and results were well correlated with drilled profiles in Central TX. Information is provided for the VES array assembly, field measuring procedure and interpretation of sounding results. Previously, device was used in Astrakhan area, Russia for estimation of the groundwater table and salinity layers in the soil profiles. The method of self-potential was used to estimate subsurface water flux directions and intensities through the measured variation in electrical potential on the soil surface and direct potable wells placement in Kiev, Ukraine and Dmitrov, Russia.

Introduction

Electrical methods are intensively used by geophysicists for evaluation of deep subsurface. Measurements of the electrical conductivity or resistivity have been applied for soil salinity surveys in situ for many years (Austin and Rhoades 1979; Rhoades and Ingvalson 1971; Rhoades et al. 1990). The most common method is the electrical profiling using four-electrode probes in Wenner configuration. The probes are applied on the soil surface as well as in bore-hole logging (Rhoades and Schilfgaarde 1976; Halvorson and Rhoades 1976; Rhoades 1979). Recently other electrical geophysical methods such as electromagnetic induction (EM) and ground penetrating radar (GPR) become increasingly popular. The methods are still applied preferentially on saline irrigated areas. Some successful applications of the methods were reported on accessing quality of forest soils (McBride, Gordon, and Shrive 1990), mapping water flow paths (Freeland et al. 1997), finding perched water locations (Allred et al. 2010), and outlining permafrost layers (Arcone et al. 1998). Despite the promising applications, methods of four-electrode profiling, EM, and GPR have some drawbacks when used for shallow soil profiles. Methods of EM and four-electrode probe cannot directly measure different resistivities or conductivities of soil horizons and provide only average or bulk electrical conductivity of the soil profile (D. L. Corwin and Rhoades 1984). GPR evaluates profile differentiation in soil electrical conductivity, but its

application is limited on soils with high conductivity (salty soils, clay soils). GPR is also not easily modified for shallow subsurface measurements (Liner and Liner 1997).

Although the method of vertical electrical sounding (VES) is very popular in conventional geophysical studies, such as gas, oil, and coal exploration (Verma and Bandyopadhyay 1983), it is rarely used in shallow subsurface studies. Vertical electrical sounding was applied to estimate hydraulic conductivity (Mazac et al. 1990) and texture (Banton, Seguin, and Cimon 1997) of the stratified soils and sediments. Barker (1990) applied VES to a landfill outlining at a 40-m depth.

In search for potable water we need to know stratigraphy of shallow subsurface (generally no more than 60-100 m) and be able to distinguish between dry and saturated subsurface layers. Those layers usually have contrasting resistivities and VES method (1D) can be easily adapted to such studies (Golovko, Pozdnyakova, and Pozdnyakov 2011; Pozdnyakova, Pozdnyakov, and Zhang 2001; Smernikov, Pozdnyakov, and Shein 2008).

The SP method was used by Fox as early as 1830 on sulphide veins in a Cornish mine, but the systematic use of the SP and electrical resistivity methods in conventional geophysics dates from about 1920 (Parasnis 1997). The SP method is based on measuring the natural potential differences, which generally exist between any two points on the ground. These potentials are associated with electrical currents in the soil. Large potentials are generally observed over sulphide and graphite ore bodies, graphitic shale, magnetite, galena, and other electronically highly conducting minerals (usually negative). However, SP anomalies are greatly affected by local geological and topographical conditions. These effects are considered in exploration geophysics as “noise”. The electrical potential anomalies over the highly conducting rock are usually overcome these environmental “noise”, thus, the natural electrical potentials existing in soils are usually not considered in conventional geophysics.

In soil and hydrological studies researchers are especially interested in the measurement of such “noise” electrical potentials created in soils due to soil-forming process and water/ion movements. The electrical potentials in soils, clays, marls, and other water-saturated and unsaturated sediments can be explained by such phenomena as ionic layers, electro-filtration, pH differences, and electro-osmosis (A. Pozdnyakov, Fedotov, and Pozdnyakova 2003). Soil-forming processes can create electrically variable horizons in soil profiles, thus electrical potential differences measured between soil horizons can be used to study soil forming processes and soil genesis (McKenzie and Erickson 1954; Pozdnyakov 2008). The most popular environmental and engineering application of self-potential method is to study subsurface water movement (Furman, Warrick, and Ferrer 2002; Butler 2007; Tchoumkam et al. 2010). Measurements of electro-filtration potentials or streaming potentials have been used in Russia to detect water leakage spots on the submerged slopes of earth dams. Method of self-potential in addition to EC mapping and vertical electrical sounding/ imaging (VES) can aid in archaeological and civil engineering projects (Golovko and Pozdnyakov 2010; Pozdnyakova, Pozdnyakov, and Zhang 2001; Smernikov, Pozdnyakov, and Shein 2008).

Thus, VES and SP methods complement each other and have been used in shallow hydrological studies to detect stratigraphy of subsoil and bedrock, depth to groundwater table and subsurface water movements. All this information can be obtained directly from soil surface with those non-destructive electrical geophysical methods.

Materials and Methods

Landviser, LLC have developed a new model of our popular soil resistivity meter, LandMapper ERM-02, with combined capabilities to use in VES and SP studies (Golovko, Pozdnyakov, and Pozdnyakova 2010). ERM-02 can automatically output EC or ER, accepts four-electrode probes of any configuration, including dipole-dipole and rectangular probes, reaches down to 25 m depth in most soils

and stores up to 999 resistivity values in non-volatile memory (Fig. 1) (Loke, Chambers, and Kuras 2011). Also, ERM-02 model can be used to measure natural electrical potentials in soils, plants and other media with two non-polarizing electrodes. This device was tested in 2011 by Water for All International, non-profit organization on their test farm in San Angelo, TX.



Figure 1. Illustration of LandMapper use with 1D VES array.

The vertical electrical sounding and electrical profiling methods are based on the four-electrode principle as shown in Fig. 1. The electrical current (I) is applied to A and B electrodes and the potential (ΔU) is measured between M and N electrodes. The bulk soil electrical resistivity (ER) is calculated with

$$ER = K \frac{\Delta U}{I} \quad [1]$$

where K is the geometric factor.

Some uncertainties exist in the soil literature about the calculation of K and estimation of the measuring depth with different arrays (Banton, Seguin, and Cimon 1997; Kirkham and Taylor 1949). As implied in conventional geophysics, the depth of penetration of electrical field in the media is influenced by the array geometry as well as electrical conductivity and layer organization of the media (Barker 1989; Parasnis 1997). Therefore, the depth of penetration cannot be precisely derived from the distances between the electrodes in an array. Theoretical derivations and practical tests have shown that the approximate penetration depth can be considered as 1/6 of $[AB]$ for the arrays of Schlumberger and Wenner types used on wide range of soils and grounds (Barker 1989; Banton, Seguin, and Cimon 1997; A. I. Pozdnyakov, Pozdnyakova, and Pozdnyakova 1996).

While the depth of penetration for an array varies for the different soils around 1/6 of [AB], the geometric factor (K) can be precisely derived from the array geometry based on the law of electrical field distribution. Using the Laplace equation in polar coordinates, Keller and Frischknecht (1970) derived the electrical potential functions around the source (A and B) and measuring (M and N) electrodes. The geometric factor K can be obtained for four-electrode array of AMNB configuration as

$$K = \frac{2\pi}{\frac{1}{[AM]} - \frac{1}{[BM]} - \frac{1}{[AN]} + \frac{1}{[BN]}} \quad [2]$$

where [AM], [BM], [AN], and [BN] are the distances (m) between the respective electrodes. For central-symmetric array, when [AM]=[BN] and [BM]=[AN], Eq. [2] can be simplified to

$$K = \pi \frac{[AM][AN]}{[MN]} \quad [3]$$

The VES array consists of a series of the electrode combinations AMNB with gradually increasing distances among the electrodes for consequent combinations (Fig. 1). The depth of sounding increases with the distance between A and B electrodes. The K factors for the combinations are calculated with Eq. [3] and used to obtain electrical resistivity from measured electrical potential and current using Eq. [1]. The result of VES measurements with central-symmetric arrays is apparent (bulk) electrical resistivity as a function of half of the distance between the current electrodes, i.e. $ER=f(AB/2)$ (Beck 1981).

For this study we constructed a simple manual VES array with maximal distance between [AB/2] electrodes equal to 30 m, which roughly could reach to about 10 m depth. Test measurements were conducted, then a borehole was drilled to the water layer and subsurface stratigraphy was examined in the core. Landviser, LLC has developed and provided to the customers a simple Excel spreadsheet with directions on how to conduct VES measurements (Fig. 2). After raw resistivity data entered in the spreadsheet, the Excel table automatically calculates apparent resistivities for the respective AB/2 and $ER=f(AB/2)$ is plotted (Fig. 3). Thus, the extreme values of resistivity are instantly visible and approximate depth to bedrock could be identified.

For more accurate analysis, the relationship between ER and $AB/2$ can be converted into a relationship between electrical resistivity and actual soil depth through a computer interpretation. Pozdnyakov et al. (1996) developed a user-interactive program for soil VES interpretations based on an updated R-function (iVEZ, Landviser LLC, www.landviser.com). This program is especially useful for more accurate analysis of shallow soil layers with diffuse boundaries. Also, there are other 1D VES interpretation software is available, such as IPI2win (V.A. Shevnin and I.N.Modin) and RES1D (Geotomo Software, Malaysia, www.geoelectrical.com). The electrical resistivity measured with the VES method is shown to be related with salinity, texture and structure, porosity, and bulk density of the soil (Pozdnyakov et al., 1996; Banton et al., 1997). The interpolated VES resistivity profiles can provide useful information on the geological structures, soil properties, and hydrological conditions in a study area.

Previously we examined soil profiles with highly variable electrical resistivities caused by water re-distribution due to groundwater fluctuations. The VES method was applied to evaluate of saline layers and groundwater depths in the alluvial soils of delta Volga, Russia. The VES method was also utilized for in-situ monitoring of soil defrosting and drying processes on cultivated Alfisols and Histosols in Moscow area, Russia. The profile distributions of stones in soils of Crimea Peninsula, Russia were successfully investigated with the VES method.

Big Probe for Manual VES

Max Depth ~ 20 m, experimental

Manual field VES-1D procedure:

1. Completely unwind both red wires from the probe. The wires have A and B electrodes at the ends.
2. Ground central black MN electrodes. Completely spread red wires with A and B electrodes in straight line and insert electrodes in soil => AB/2=30 m
3. Connect black wires to MN terminals of Landmapper.
4. Connect red wires to AB terminals of Landmapper
5. Take ER-reading at K0=1 and AB electrodes at 30.0 m marks (completely spread wires)
6. Move A and B electrodes to 25 m marks at both sides - take ER-reading
7. Continue until all ER-readings are taken (last AB/2=3 m)
8. Field ER-readings can be stored in LandMapper memory or recorded manually
9. The recorded ER-values needs to be multiplied on appropriate K-coefficients - see below

LandMapper automatically measured voltage, current and calculates ER (electrical resistivity) by formula below
 $ER = K * U/I$
U is potential measured between M and N
I is current measured between A and B
ERM-02 uses K0=1 stored permanently or K1-K9 (user changeable)

Note: if ER values measured on deep AB/2 (#1-5) are too small, switch to K1=10, or even K2=100, don't forget to record which K-taken you used, and table on second page will re-calculate ER automatically.

AB distances, m	Measurement from deep to shallow					[MN] = 2 m, fixed for all 15 measurements							
	#	AB/2, m	K-set	K-taken	ER-1	ER-2	ER-3	ER-3	ER-5	ER-6	ER-7	ER-8	ER-9
[A1B1] = 60	1	30	K0=1	1	0.1	4.05							
[A2B2] = 50	2	25	K0=1	1	3	5.3							
[A3B3] = 40	3	20	K0=1	1	4	57							
[A4B4] = 36	4	18	K0=1	1	10	4.96							
[A5B5] = 32	5	16	K0=1	1	51	5.58							
[A6B6] = 28	6	14	K0=1	1	45	2347							
[A7B7] = 24	7	12	K0=1	1	7	28.84							
[A8B8] = 20	8	10	K0=1	1	5	8.19							
[A9B9] = 18	9	9	K0=1	1	12	9.6							
[A10B10] = 16	10	8	K0=1	1	25	6.34							
[A11B11] = 14	11	7	K0=1	1	30	1639							
[A12B12] = 12	12	6	K0=1	1	50	5.67							
[A13B13] = 10	13	5	K0=1	1	100	2790							
[A14B14] = 8	14	4	K0=1	1	26	52.97							
[A15B15] = 6	15	3	K0=1	1	60	10.56							

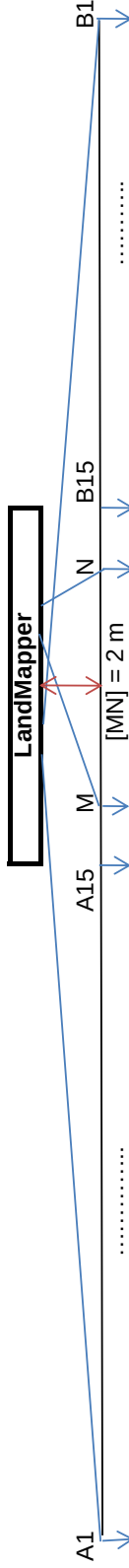


Figure 2: Example of spreadsheet with entered field electrical resistivity data.

A few mock data entered as an example.

$$K\text{-multiply} = \frac{[AM] * [AN]}{[MN]}$$

K for central-symmetric arrays

In all geophysical procedures distance always in **Meters**.

This table recalculates actual ER values suitable for interpretation with iVEZ if measured field ER entered in the table above.

AB distances, m	Measurement from deep to shallow				Recalculate ER for VES interpretation								
	#	AB/2, m	K-multiply	K-taken	ER-1	ER-2	ER-3	ER-3	ER-5	ER-6	ER-7	ER-8	ER-9
[A1B1] = 60	1	30	1412.1	1	141.2149	5719.204		0	0	0	0	0	0
[A2B2] = 50	2	25	980.2	1	2940.538	5194.95		0	0	0	0	0	0
[A3B3] = 40	3	20	626.7	1	2506.997	35724.7		0	0	0	0	0	0
[A4B4] = 36	4	18	507.4	1	5073.684	2516.547		0	0	0	0	0	0
[A5B5] = 32	5	16	400.6	1	20428.25	2235.091		0	0	0	0	0	0
[A6B6] = 28	6	14	306.3	1	13783.77	718900.2		0	0	0	0	0	0
[A7B7] = 24	7	12	224.6	1	1572.371	6478.168		0	0	0	0	0	0
[A8B8] = 20	8	10	155.5	1	777.546	1273.62		0	0	0	0	0	0
[A9B9] = 18	9	9	125.7	1	1507.968	1206.374		0	0	0	0	0	0
[A10B10] = 16	10	8	99.0	1	2474.01	627.4089		0	0	0	0	0	0
[A11B11] = 14	11	7	75.4	1	2261.952	123578		0	0	0	0	0	0
[A12B12] = 12	12	6	55.0	1	2748.9	311.7253		0	0	0	0	0	0
[A13B13] = 10	13	5	37.7	1	3769.92	105180.8		0	0	0	0	0	0
[A14B14] = 8	14	4	23.6	1	612.612	1248.079		0	0	0	0	0	0
[A15B15] = 6	15	3	12.6	1	753.984	132.7012		0	0	0	0	0	0

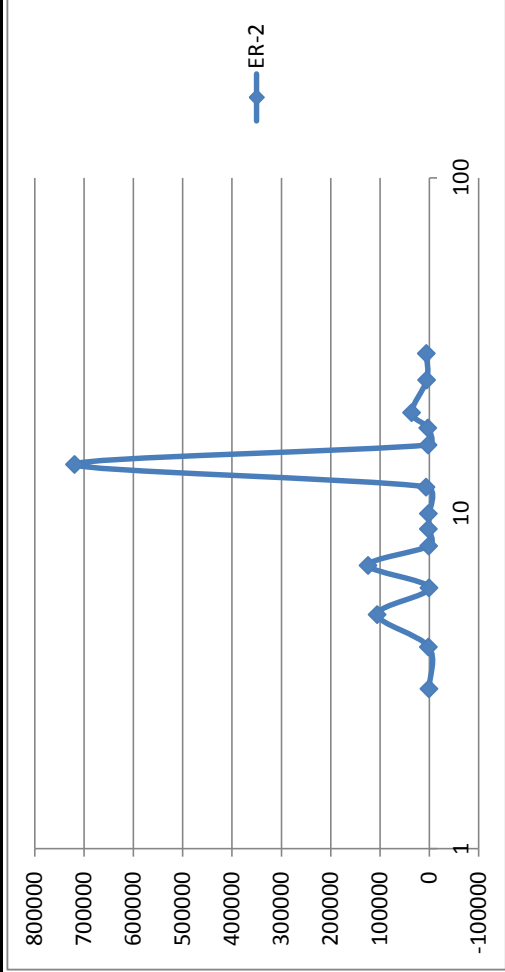


Figure 3: Example of spreadsheet with recalculated and plotted apparent electrical resistivity data.

LandMapper ERM-02, in addition to electrical conductivity and resistivity measurements also allows non-invasively measure natural electrical potentials in soils and plants when two special non-polarizing electrodes are connected to MN terminals (Figure 4). The usage of non-polarizing electrodes is mandatory when the SP method is applied in soil and environmental studies. The non-polarizing electrode consists of a metal element immersed in a solution of salt of the same metal with a porous membrane between the solution and the soil (R. F. Corwin 1990).

To measure small electrical potential differences in soils accurately, in addition to non-polarizing electrodes, the measuring device should be modified and as such should have isolated connectors and high internal impedance. Most leading geophysical resistivity instruments, such as ABEM SAS, Syskal, and Sting provide such connections, but coupling electrodes are bulky, leaky and generally not useable in plant/soil studies. LandMapper ERM-02 with self-potential measuring capability and easy coupling with medical-grade non-polarizing electrodes.

Detail description of SP method with LandMapper ERM-02 and case studies were presented in (Golovko and Pozdnyakov 2010).



Figure 4: Solid-state Ag-AgCl non-polarizing electrodes (In-Vivo Metrics, CA).

Results and Discussion

Stone content evaluation of shallow bedrock soil

Layers of different electrical resistivities are observed in stony or rocky soils. The resistivity of rocks or stones is much higher (about 10^4 - 10^{12} ohm m) than the resistivity of soil horizons with any texture (Parasnis, 1997). Based on the principle the VES was applied to locate depths of stone layers in soils of Crimea Peninsula, Russia (Lithic Xerorthent). Most of the soils could be well characterized by three-layer profile. The top layer (I) had the smallest stone content (0.22 - 0.41 cm³ cm⁻³) with electrical resistivity about 80 ohm m (Fig. 5). The middle layer (II) had the highest stone content (>50 cm³ cm⁻³) and electrical resistivity as high as 450 ohm m. Stones in layer II were cemented by carbonate marine deposits. The bottom layer (III) was not always presented in soil profiles. In some profiles layer III was outlined as having lower resistivity (40-200 ohm m) than layer II. This indicated decrease in stone content in the bottom layer compared with layer II and a loose organization of stones in the bottom layer. The approximate stone content of soil profiles was evaluated observing VES profiles. For example in Fig. 5, the stone content in the soil profiles decreased in a row VES 3-7-2-16-10. The stone content estimations based on VES profiles were verified by the direct measurement of the relative volume of stones in the soil layers. Using the VES interpretation data and the direct measurements of the stone content in 16 soil profiles, we developed a relationship between electrical resistivity measured by the VES and soil stone content. The power relationship was obtained for $1/\text{stone content}$ vs. electrical resistivity. The relationship is shown in Fig. 6 and has a correlation coefficient (r) of 0.863.

The relationship is useful for a quick judgment about stone content in different soil layers. We developed a rough scale for evaluation of stone contents in soils (Table 1). With the scale, the territory about 400 acres was mapped with the VES and four-electrode profiling methods, the stone content and

texture of soils were evaluated, and special recommendations were developed for usage these otherwise unproductive soils under the orchards.

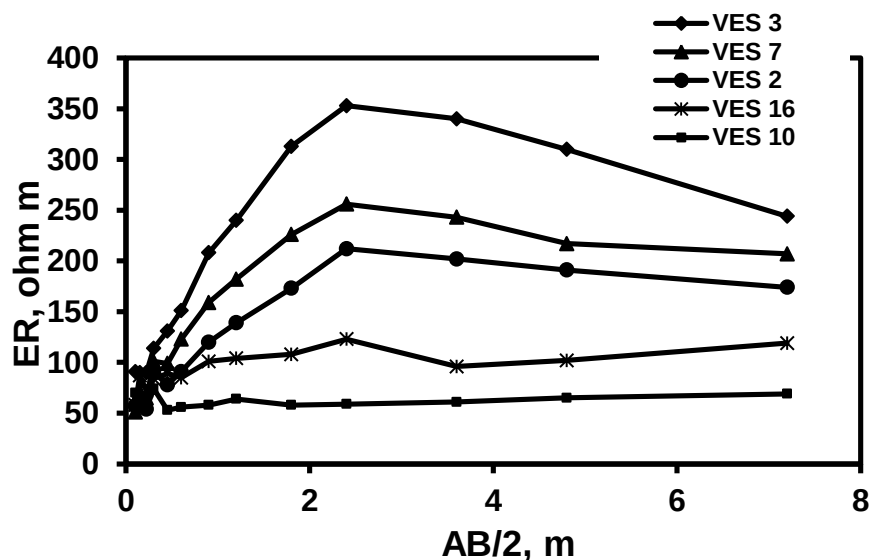


Figure 5: Profile distributions of electrical resistivity measured by VES method in stony soils of Crimea Peninsula, Ukraine.

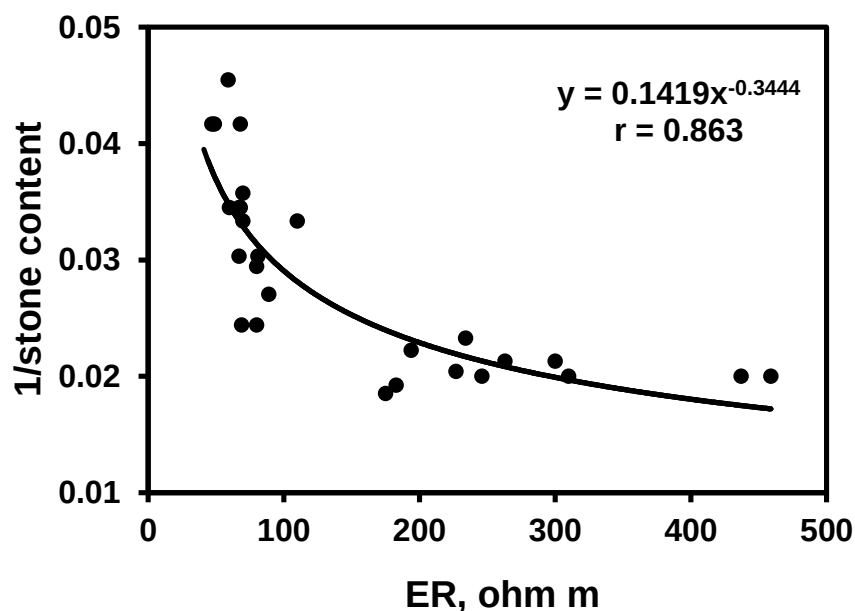


Figure 6: The relationship between the electrical resistivity measured in-situ by VES method and the volumetric stone content.

Table 1. Scale for evaluation of soil stone content by electrical resistivity measured with VES

Stone content	Electrical resistivity
$\text{cm}^3 \text{ cm}^{-3}$	ohm m
<0.2	50-80
0.2-0.4	80-120
0.4-0.6	120-150

Estimation of saline layers and groundwater tables in alluvial soils of Russia

One of the applications of the electrical resistivity or conductivity methods is to map and evaluate soil salinity. The VES was used to outline the layers with different salinity in alluvial soil profiles at delta Volga (Halaquaepts). A typical profile of the alluvial soil consisted of thin layers of silt, clay, and sand. Soil forming processes did not develop highly pronounced horizons in these soils as compared with Spodosols, Alfisols, and Ultisols. Water and salt content distributions within the soil profile are the main properties causing considerable variations in electrical resistivity.

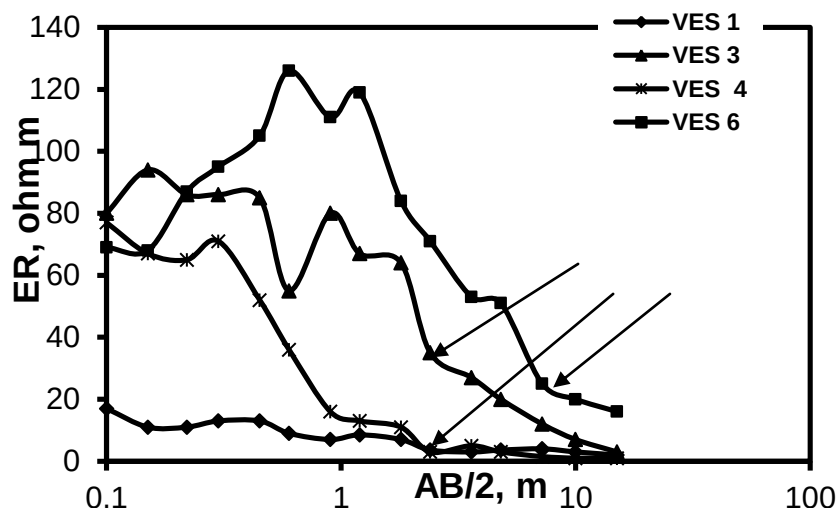


Figure 7. Electrical resistivity distributions in Halaquaepts profile indicated the groundwater depth. Arrows indicate groundwater table.

Since the evaporation in the area was about five times higher than the precipitation, the water content and salt distributions were determined mainly by the saline groundwater. The soil profile was divided into the top unsaturated layer with high resistivity and the bottom layer saturated by saline groundwater with low resistivity. Considering large differences in electrical resistivity between the unsaturated and saturated zones, the VES method was applied to detect the saline groundwater level. The approximate location of the groundwater table was estimated by a visual inspection of the VES curve. The $AB/2$ value with the sharp change to the low resistivity (3-20 ohm m) was selected from each VES profile and multiplied by an empirical coefficient (Fig. 7). This coefficient was estimated as 0.32 for the investigated soils and varied from 0.28 to 0.34 for other soil types (Barker, 1989). For example, for VES 6 the $AB/2$ with such sharp change is 7.2 m and the groundwater table is estimated as $7.2 \times 0.32 = 2.3$ m (Fig. 7). In some cases (Fig. 7, VES 3) it was difficult to visually determinate where the VES curve has a sharp change in electrical resistivity. Nevertheless, with the computer interpretation of the VES data we could determine the changes more accurately. The results of the computer interpretation of the VES data were compared with the real groundwater tables measured in bore-holes and the relative errors of the VES estimation varied from 3 to 11% as shown in Table 2.

The differentiation of salinity in the unsaturated zone of the soil profiles was revealed by small fluctuations of electrical resistivity in upper part of the VES profiles (Fig. 7). We thoroughly interpreted the VES results to estimate the layers with different resistivities for 12 soil profiles. The total salt content was measured in soil samples collected from the layers of soil profiles as shown in Table 3 (columns 1 and 2) for one example profile. The VES method outlined three layers with different resistivities for the same soil profile (Table 3, columns 3 and 4). In the column 5 a weighted averages for the outlined soil layers were recalculated from the total salt contents in column 2 (Table 3).

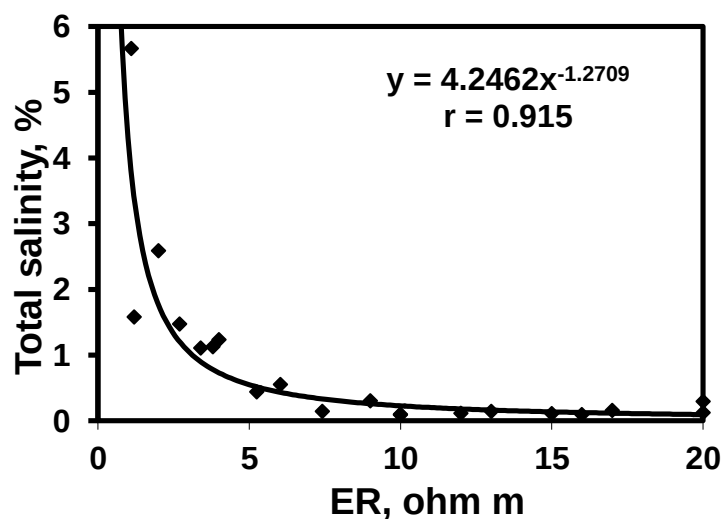
Table 2. Estimation of groundwater table with the vertical electrical sounding.

Case number	Groundwater table		Relative estimation error
	Real (bore hole)	Estimated (VES)	
	m		%
1	2.19	2.37	7
2	1.15	1.29	11
3	2.47	2.55	3
4	2.38	2.25	5
5	1.60	1.53	5
6	1.17	1.34	8
7	1.32	1.22	8
8	1.38	1.24	10
Mean			8

Table 3. Illustration to evaluation of soil salinity with the vertical electrical sounding.

Depth	Total salinity	Results of interpretation		Recalculated salinity
		Layer depth	ER	
— m —	— % —	— m —	— ohm m —	— % —
0-0.02	0.092	0-0.17	98	0.074
0.02-0.05	0.087			
0.05-0.20	0.068	0.17-0.74	15	0.095
0.20-0.40	0.07			
0.40-0.70	0.112			
0.70-1	0.117	0.74-2.55	12	0.117

Data of recalculated total soil salinity and VES electrical resistivity were combined from the layers of all 12 soil profiles to obtain a relationship between electrical resistivity and total salt content (Fig. 8). The correlation coefficient (r) was calculated for the linearized power relationship as 0.915. Data for the soil layers with resistivity higher than 20 ohm m were excluded from the correlation analyses, because they were mainly corresponded to non-saline and very low saline soils with the total salt content less than 0.3%. For quick delineation and estimation of salinity in a soil profile we can consider that a resistivity of 10-20 ohm m corresponds to a total salt content between 0.3 and 0.5% and a resistivity <3 ohm m indicate that the total salt content in soil is >1%.

**Figure 8:** The relationship between the electrical resistivity measured in-situ by VES method and the total salt content in soils of delta Volga, Russia.

Estimation of drying depth in peat soil

Usually water content of cultivated peat soils is close to the field capacity during the whole growing season. The other soil properties, such as bulk density, texture, and ash content, that might influence electrical resistivity, are also practically uniform in the soil profile. Therefore, a typical electrical resistivity distribution in the profile of cultivated peat soil (Hemic Haplosaprist) is uniform and about 100 ohm m during the years with average precipitation (Fig. 9, VES for 4 July, 1994). In extremely dry years as in the summer of 1995 in Moscow area, Russia, the upper 50-cm layer of peat soil dried almost to the wilting point, causing an increase of electrical resistivity up to 1,500 ohm m (Fig. 9). The drying depth was precisely determined with the VES interpretation and verified with direct water content measurements.

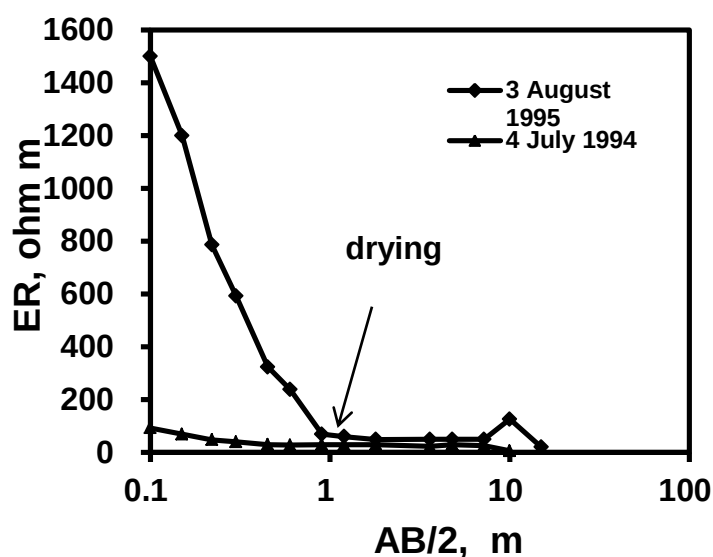


Figure 9: Determination of drying depth on the cultivated Hemic Haplosaprist by the VES measurements. Moscow area, Russia.

Adding Self-Potential mapping to VES for complex hydrology surveys

Method of SP is complimentary to electrical resistivity mapping and VES. In Kiev-Pechersk Lavra, the problem area was located on a hill far above the groundwater table. Therefore, the problem was attributable to temporary subsurface water fluxes fed by precipitation. Excess water accumulated in subsurface in spring because of snow melting and in summer during intensive rainfalls. Due to the hill topography, water could accumulate in soil covering the whole territory of Upper Lavra and then flow into the Patriarch Garden as shallow subsurface fluxes (Fig. 10). We used the VES and EP methods to investigate the properties of water-bearing and waterproof layers essential for the development of the subsurface water fluxes. The directions and intensities of the fluxes were evaluated with the SP method.

The VES and EP methods revealed complex stratification of the hill slope in the Patriarch garden near The Church of Holy Cross Elevation (Fig. 10). All the VES curves revealed three-layer soil profile with apparent $ER_1 > ER_2 < ER_3$ (Fig. 11). The top layer was represented by the eroded Chernozem of coarse textures with electrical resistivity (ER_1) about 125 ohm m for loamy sand and about 50 ohm m for sandy clay loam. The second layer was a thick clay layer (7 m) with low electrical resistivity (ER_2) from 2 to 16 ohm m. The clay was saturated and gleyed in some places, which was indicated by 2 ohm m resistivity. The third layer with the resistivity (ER_3) about 2000 ohm m was horizontally deposited sandstone. The results of sounding were verified with boring at the same 12 locations (Fig. 10).

Soil thickness was only about 20 cm on the top of the hill, but it increased toward the gallery gate up to 2 m (Fig. 10, VES 1 and 4). Soil thickness increased up to 2.5 m from the gallery gate further downhill (Fig. 10, VES 17). Along the same direction the texture in surface soil layer changed from loamy sand (VES 4) to clay loam and clay (VES 17 and 21). That change in the texture of the first layer was reflected with the decrease in electrical resistivity (ER_1) measured by the EP along the Route 1 (Fig. 10). The electrical resistivity was about 100 ohm m for peaty sand at the top part of the Route 1 (VES

4), then decreased to 65 ohm m for sandy loam (VES 12), and further decreased to 35 ohm m on the surface and to 7 ohm m on the 0.5 m depth for clays near the tier wall (VES 21).

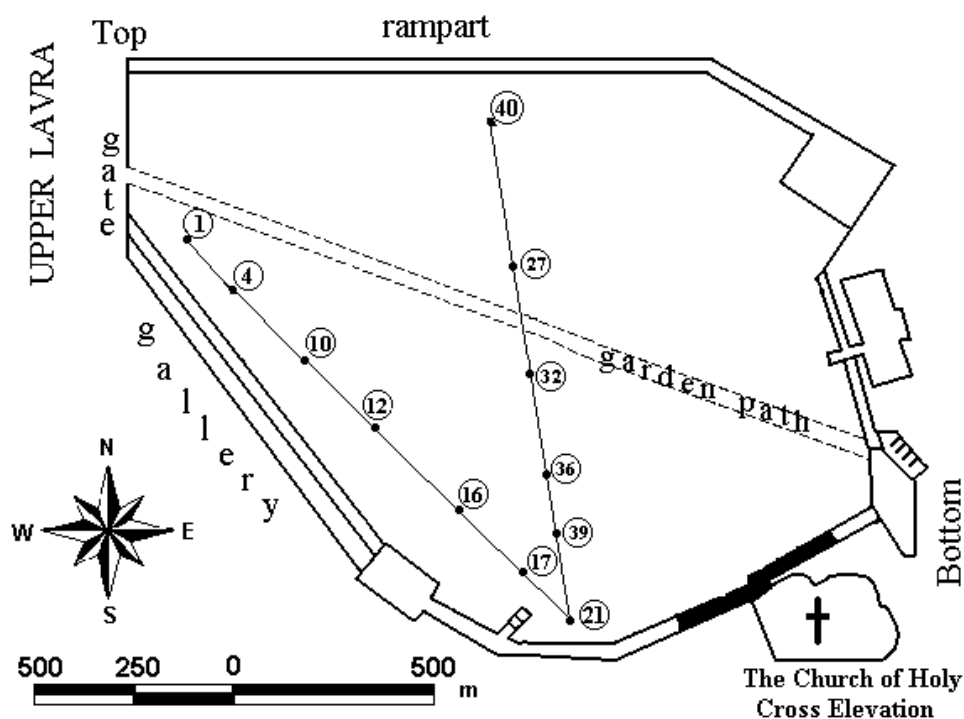


Figure 10: Scheme of the investigated area within Patriarch garden, Kiev, Ukraine: the numbers in circles show VES and boring locations, lines are routes for EP investigation (Route 1 from VES 1 to VES 21 and Route 2 from VES 27 to VES 21), and the shaded area near the church is a concrete wall. Height difference between top and bottom of the hill is about 28 m.

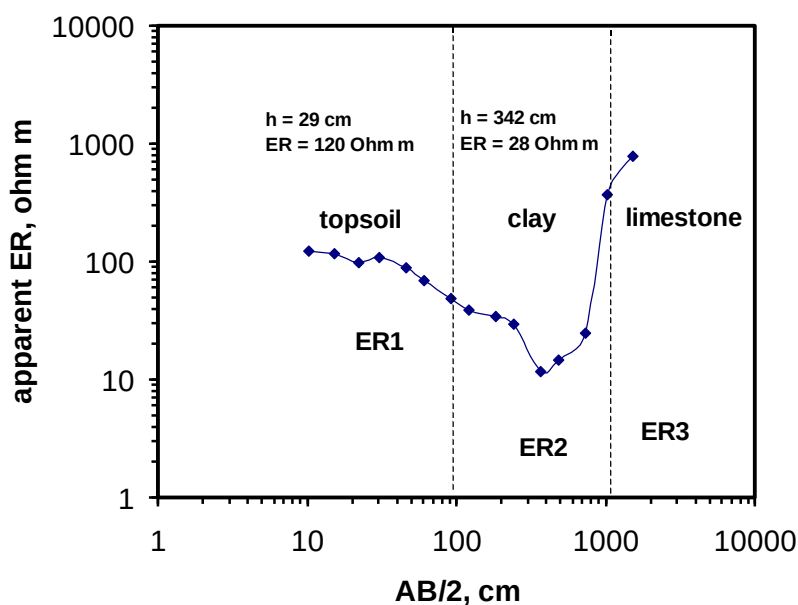


Figure 11: Typical distribution (VES 4) of electrical resistivity within soil profile measured by VES in Patriarch garden, Kiev, Ukraine. Thickness (h) and resistivity (ER) for the layers are derived through computer interpretation.

The thickness of clay layer decreased from 8 to 2 m along the line from the top of the hill to the tier wall (Fig. 10). The low and almost constant electrical resistivity (8 ohm) for the AB/2 from 3.6 to 7.2 m for the second layer shown that clay did not bear any intrusions of sand or sandy loam, which was verified with boring (Fig. 11). Therefore, water flow inside second layer was impossible. The water flow could be formed only in the topsoil over the layer of waterproof clay.

Although undetectable on the surface, three gullies were revealed in the second layer of waterproof clays by the VES and EP methods. The subsurface water flow could be formed in such gullies. The first gully was detected near the gallery wall along the Route 1. The subsurface water flowed in the gully from the Upper Lavra downhill. Waterproof clay occurred on the surface about 6 m apart from the gallery wall (VES 12) causing bend of the subsurface water flow toward the Church of Holly Cross Elevation. The second gully formed along the garden path, which was indicated by non-decomposed plant debris deposited here. Narrow (about 17 m) band of raw organic litter mixed with sand of 30-cm depth was formed along the garden path (Fig. 10, VES 1, 27, and 32). This indication of starting wetland formation on such steep slope designated that water saturation had occurred here for at least ten years. The second gully was also directed toward the Church of Holly Cross Elevation. The third gully was formed near the rampart and separated from the first and second gullies. Thus, the stratification of the investigated area with the EP and VES methods shown three possible ways for water flow in the soil: (1) near the gallery, (2) along the garden path, and (3) near the rampart.

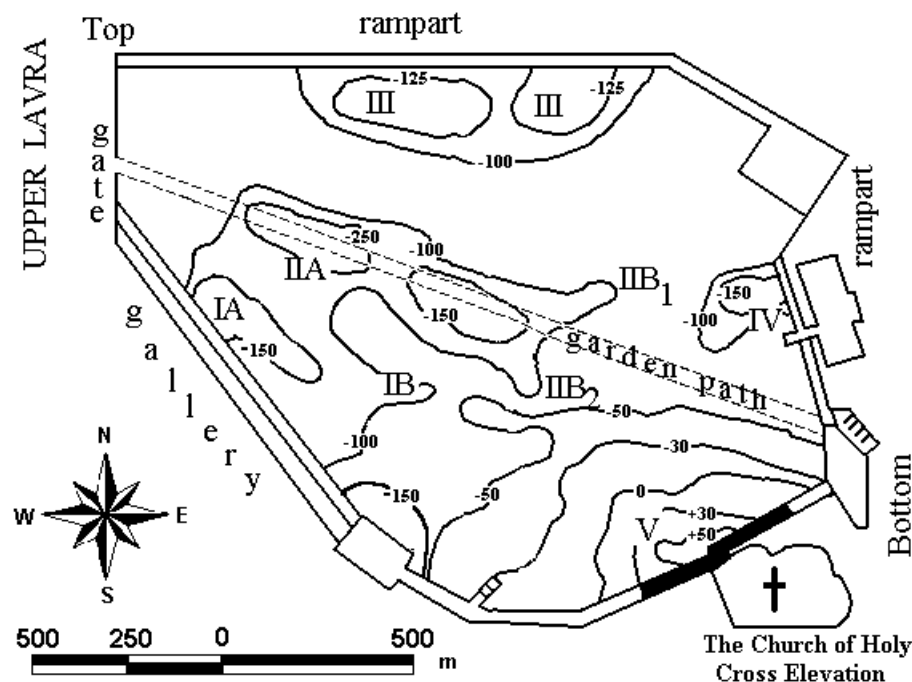


Figure 13: Results of the SP measurements within Patriarch garden, Kiev, Ukraine. The curves are iso-potential lines. The numbers (I and II) and letters (A and B) indicate two fluxes with water flow from A to B areas; numbers III and IV indicate zones of infiltration; and number V shows seepage.

The method of self-potential was used to estimate water flux directions and intensities through the measured variation in electrical potential on the soil surface. An iso-potential map (Fig. 13) was developed with the method on a 5 x 10 m grid; 299 locations were measured with five replications.

Three major iso-potential areas detected from Fig. 13. Two areas with negative potentials were formed near rampart and along the gallery (including the garden path) and indicated the areas of water infiltration into the soil and development of groundwater flow (Fig. 13, I, II, and III). The third area with positive potentials outlined the seepage zone near The Church of Holy Cross Elevation (Fig. 13, V). The most negative potentials (-250 mV) along the garden path indicated the most intensive subsurface water flow in this area (Fig. 13, IIA, IIB₁, and IIB₂). The -250 -mV iso-potential area developed in surface peaty sand with electrical resistivity (ER_1) about 170 ohm m. The less negative potentials (-150 mV) and, therefore, the less intensive water flow occurred near the gallery (Fig. 13, IA and IB). The same negative potential areas were detected in the middle of the garden path and near the rampart (Fig. 13, IIIA, IIIB, and IV). The seepage area was outlined by the 0 -mV iso-potential near The Church of Holy Cross Elevation (Fig. 13, V). The seepage area was enriched with clay material having electrical resistivity about 5 ohm m. The percentage of clay in the soil increased toward the corner of the church along with the electrical potential.

The directions of water flow were predicted from the directions of increase in electrical potential. Three main water flux directions were detected by self-potential method: flux I near the gallery, flux II along the garden path, and flux III near the rampart (Fig. 13). Flux I was formed in thick loam sand about 1.5 m depth and characterized by a stable slow rate. Flux II could be highly intensive during rainfalls, since it was formed in coarse sand layer with a thickness about 0.3 m. Further downhill flux II was separated into two sub-fluxes (IIB₁ and IIB₂). Fluxes IB₁ and IIB₂ merged during the intensive precipitation and seeped near the church infiltrating into the holly caves. Flux III was separated from fluxes I and II and did not influence the seepage near The Church of Holy Cross Elevation.

To protect The Church of Holy Cross Elevation, the following procedures were proposed based on our geophysical exploration near the architecture memorial. First, a hedge should be constructed across the gate to the garden to prevent the surface water flow to the Patriarch Garden from the pavement of Upper Lavra. Second, a small dike should be built perpendicular to the gallery and the garden path to direct subsurface water flow from fluxes II and I into the drain system. Third, to enhance evapotranspiration, trees and bushes with the intensive transpiration ability, such as willows and poplars, should be planted along the gallery and rampart, especially in the areas indicated by low potentials. All the measures were implemented in 1990 and still provide adequate preservation for the church and the surrounding caves. The cost of the proposed measures was about one twentieth of the previous construction of concrete wall, which, nevertheless, did not solve the problem of water penetration into the caves.

Analogously with detailed survey outlined above for Kiev, Ukraine we have conducted hydrological survey with LandMapper ERM-02 on first terrace of Yahroma River near Dmitrov, Moscow area, Russia to detect best possible placement of potable water well. Two highly negative potential anomalies of electrical potential were detected, with stronger one being downhill in a location not suitable for drilling. Therefore, the less pronounced anomaly was drilled instead. The water was discovered at about five meters in coarse gravely layer imbedded between clay layers. The water flux is strong and renewal rate is enough for household pumping. The nearby wells drilled without self-potential guidance in majority produce inferior water with low recharge rates from clay layers.

Conclusions

This study provided a basis for using the vertical electrical sounding and self-potential geophysical methods in soil hydrological surveys. The conventional VES method commonly used for deep geophysical exploration was modified to make it suitable for soil studies. The specific design of the LandMapper ERM-02 instrument allows accurate measuring of the electrical resistivity distribution in soil profiles without disturbing soils and also can be used to produce self-potential maps. Electrical

resistivity (ER) or conductivity (EC) maps are generally similar to the maps of self-potentials, but using combination of those methods brings more information about infiltration and subsurface water fluxes and aid in search for placement of potable water wells, clogged drainage pipes and reclamation planning in urban and rural areas (Smernikov, Pozdnyakov, and Shein 2008; Pozdnyakova, Pozdnyakov, and Zhang 2001; Bedmar and Araguás 2002).

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